Predicting SpaceX Falcon 9 First-Stage Landing Success

IBM Applied Data Science Capstone

Author: Hong An Vo

Project repository: https://github.com/hongan151206-ai/Capstone-project

Executive Summary

- Goal: predict whether a Falcon 9 first stage lands successfully, because reuse is what lets SpaceX price a launch at ~$62M versus $165M+ for competitors.
- Methods: data from the SpaceX REST API and Wikipedia web scraping; wrangling with pandas (Class label); EDA with SQL and Seaborn/Matplotlib; Folium map and Plotly Dash dashboard; four classifiers (Logistic Regression, SVM, Decision Tree, KNN) tuned with GridSearchCV.
- Key results: 90 Falcon 9 launches, overall landing success rate 66.7%; success rose from 0.0% in 2010 to 84.2% in 2020; best model: SVM with 83.3% test accuracy.
- Conclusion: landing outcome is predictable from pre-launch features; flight experience, orbit and launch site drive success. Illustrative expected launch cost from predicted landing probability: ~$94.8M.

GitHub: https://github.com/hongan151206-ai/Capstone-project

Introduction

Project background

- Commercial space travel is growing and SpaceX leads the field thanks to reusable rockets.
- A Falcon 9 launch is advertised at ~$62M; other providers charge upwards of $165M. Most of the saving comes from reusing the first stage.

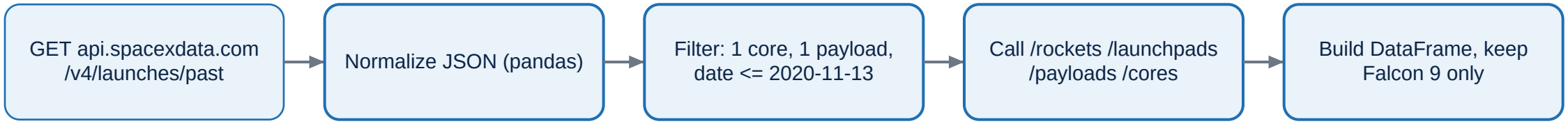
Problem statement

- We work as a data scientist for a company that wants to bid against SpaceX for a rocket launch.
- Task: determine the price of each launch by predicting whether the first stage will land successfully (and so be reused).
- Questions: what factors determine landing success? how did success change over time? which classifier predicts best?

GitHub: https://github.com/hongan151206-ai/Capstone-project

Data Collection - SpaceX API

Flowchart



- Key phrases: requests.get() -> .json() -> json_normalize() -> helper functions per endpoint
- Helpers: getBoosterVersion, getLaunchSite, getPayloadData, getCoreData
- Missing PayloadMass replaced with the column mean; Falcon 1 launches removed; FlightNumber re-indexed
- Result: dataset_part_1.csv with 90 Falcon 9 launches and 17 columns (Orbit, LaunchSite, Outcome, GridFins, Reused, Legs, ...)
- GitHub URL: https://github.com/hongan151206-ai/Capstone-project/blob/main/01_spacex_api_collection.py

Data Collection - Web Scraping

Flowchart



- Key phrases: requests.get() -> BeautifulSoup() -> find_all('table') -> parse <th>/<td> cells
- Fields: flight no., launch site, payload, mass, orbit, customer, launch outcome, booster version, landing outcome, date, time
- Helpers clean date/time, booster version, landing status and payload mass strings
- Output: spacex_web_scraped.csv; used to cross-check the API data
- GitHub URL: https://github.com/hongan151206-ai/Capstone-project/blob/main/02_web_scraping.py

Data Wrangling Methodology

- Checked missing values and data types per column.
- Counted launches per site, per orbit and each landing outcome (True/False + Ocean, RTLS, ASDS; None = not attempted).
- Label: Class = 0 for bad outcomes (False Ocean, False RTLS, False ASDS, None ASDS, None None), otherwise 1.
- Feature table: one-hot encoding of Orbit, LaunchSite, LandingPad, Serial -> dataset_part_3.csv.
- GitHub URL: https://github.com/hongan151206-ai/Capstone-project/blob/main/03_data_wrangling.py

Results on the cleaned data

Launches: 90 | success rate (Class=1): 66.7%

Launches per site:

CCSFS SLC 40: 55

KSC LC 39A: 22

VAFB SLC 4E: 13

Top orbits: GTO (27), ISS (21), VLEO (14), PO (9), LEO (7)

Landing outcomes: True ASDS (41), None None (19), True RTLS (14), False ASDS (6), True Ocean (5)

Missing values: LandingPad (26)

EDA with Data Visualization - Methodology

- Scatter plots (Seaborn catplot, hue = Class): Flight Number vs Payload Mass; Flight Number vs Launch Site; Payload Mass vs Launch Site.
- Scatter plots: Flight Number vs Orbit; Payload Mass vs Orbit - orbit-specific behaviour.
- Bar chart: success rate per orbit - which orbit types are easiest to land.
- Line chart: yearly average success rate - learning curve over time.
- Purpose: reveal how each feature relates to landing success, choose features for the models, and spot sparse categories.
- GitHub URL: https://github.com/hongan151206-ai/Capstone-project/blob/main/05_eda_visualization.py

EDA with SQL - Queries Performed

Spacex.csv loaded into SQLite table SPACEXTBL; 10 queries answer the lab questions (results on the next slides):

Q1 Unique launch sites

```
SELECT DISTINCT Launch_Site FROM SPACEXTBL
```

Q2 5 records, site begins with 'CCA'

```
SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

Q3 Total payload mass for NASA (CRS)

```
SELECT SUM(PAYLOAD_MASS__KG_) AS total_kg FROM SPACEXTBL WHERE Customer =
'NASA (CRS)'
```

Q4 Avg payload mass, booster F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) AS avg_kg FROM SPACEXTBL WHERE Booster_Version
LIKE 'F9 v1.1%'
```

Q5 First successful ground-pad landing date

```
SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground
pad)'
```

Q6 Drone-ship success, payload 4000-6000 kg

```
SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome =
'Success (drone ship)' AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ <
6000
```

Q7 Total successful vs failed missions

```
SELECT Mission_Outcome, COUNT(*) AS n FROM SPACEXTBL GROUP BY Mission_Outcome
```

Q8 Boosters carrying max payload

```
SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ =
(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)
```

Q9 2015 failed drone-ship landings (month, booster, site)

```
SELECT substr(Date,6,2) AS month, Booster_Version, Launch_Site FROM SPACEXTBL
WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date,1,4) = '2015'
```

Q10 Rank landing outcomes 2010-06-04..2017-03-20

```
SELECT Landing_Outcome, COUNT(*) AS n FROM SPACEXTBL WHERE Date BETWEEN
'2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY n DESC
```

GitHub URL: https://github.com/hongan151206-ai/Capstone-project/blob/main/04_eda_sql.py

Interactive Visual Analytics - Folium and Plotly Dash

Folium map

- Circle + label for each launch site; marker clusters coloured green (success) / red (failure) for every launch record.
- MousePosition to pick coordinates; PolyLines with haversine distance to coastline, railway, highway and nearest city.
- GitHub URL: https://github.com/hongan151206-ai/Capstone-project/blob/main/06_folium_map.py

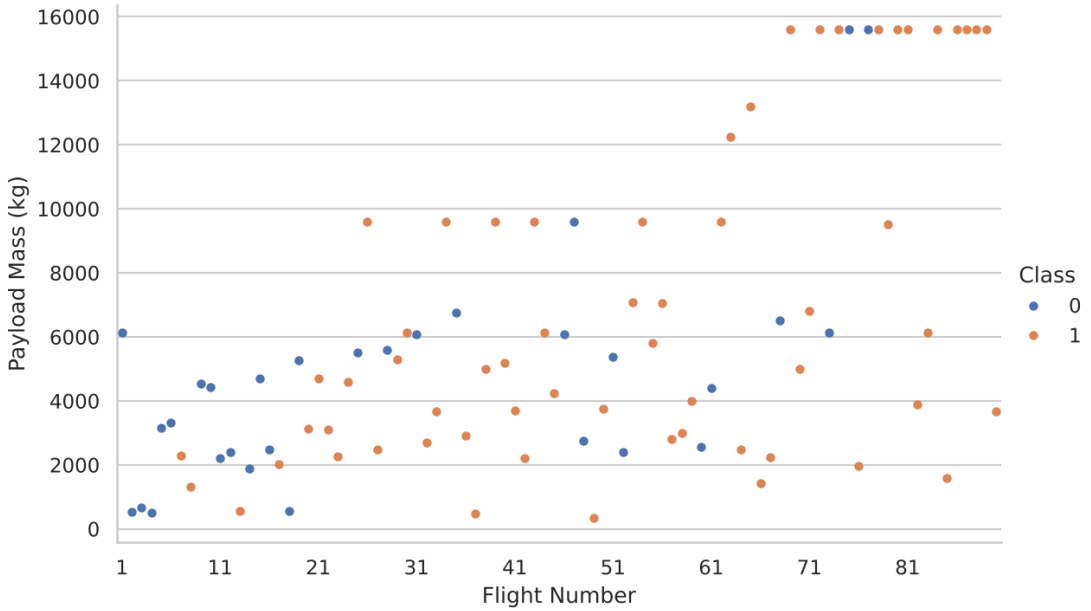
Plotly Dash app

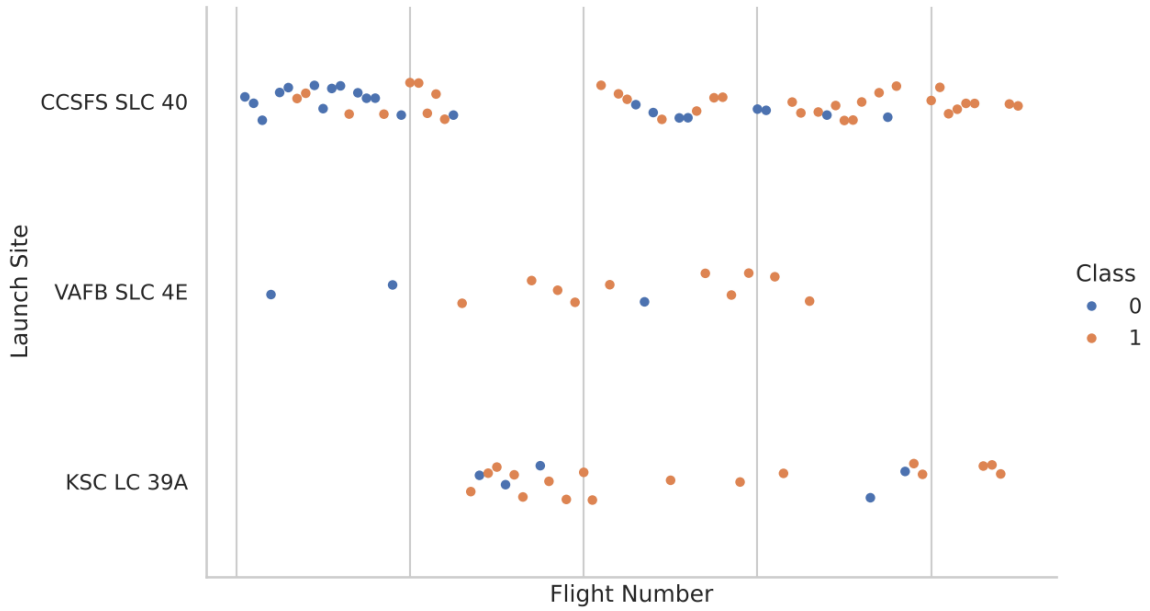
- Dropdown (all sites or one site) drives a pie chart of successful launches / success vs failure.
- Range slider filters payload mass; scatter plot of payload vs outcome coloured by booster version category.
- GitHub URL: https://github.com/hongan151206-ai/Capstone-project/blob/main/07_dash_app.py

Predictive Analysis (Classification) - Methodology

- Target: Class (1 = first stage landed). Features: dataset_part_3 (one-hot encoded), standardised with StandardScaler.
- Split: 80% train / 20% test (random_state = 2) -> 72 train, 18 test launches.
- Models: Logistic Regression, Support Vector Machine, Decision Tree, K-Nearest Neighbours.
- Tuning: GridSearchCV with 10-fold cross-validation (C/penalty; kernel/C/gamma; criterion/depth/leaf; n_neighbors/algorithm/p).
- Evaluation: test accuracy, precision, recall, F1 and confusion matrix; best model chosen on test accuracy (ties -> CV accuracy).
- GitHub URL: https://github.com/hongan151206-ai/Capstone-project/blob/main/08_predictive_analysis.py

EDA Results - Scatter Plots (Flight Number, Payload, Site)

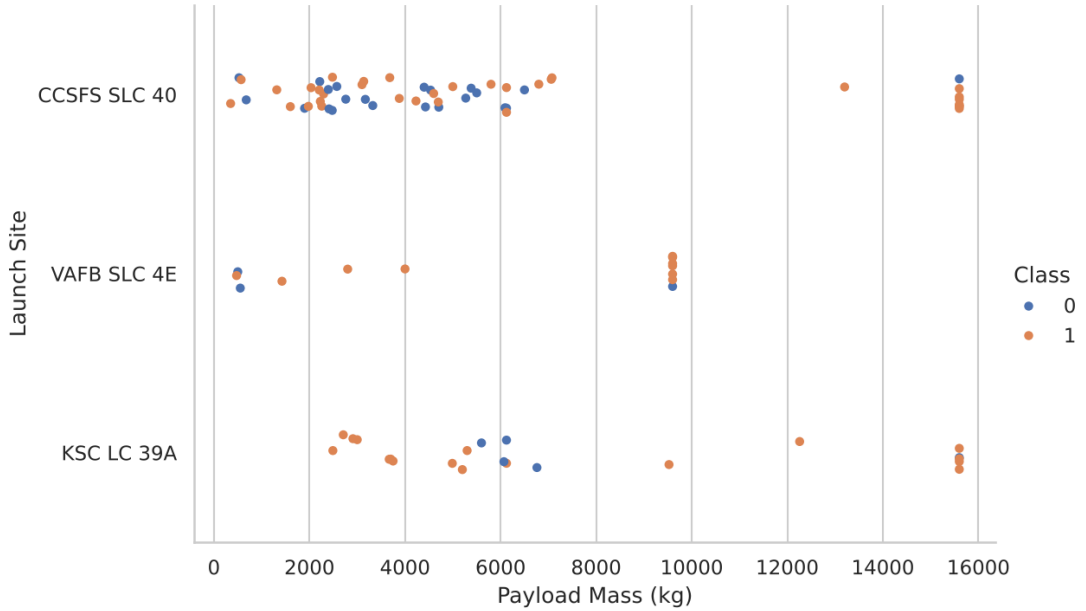


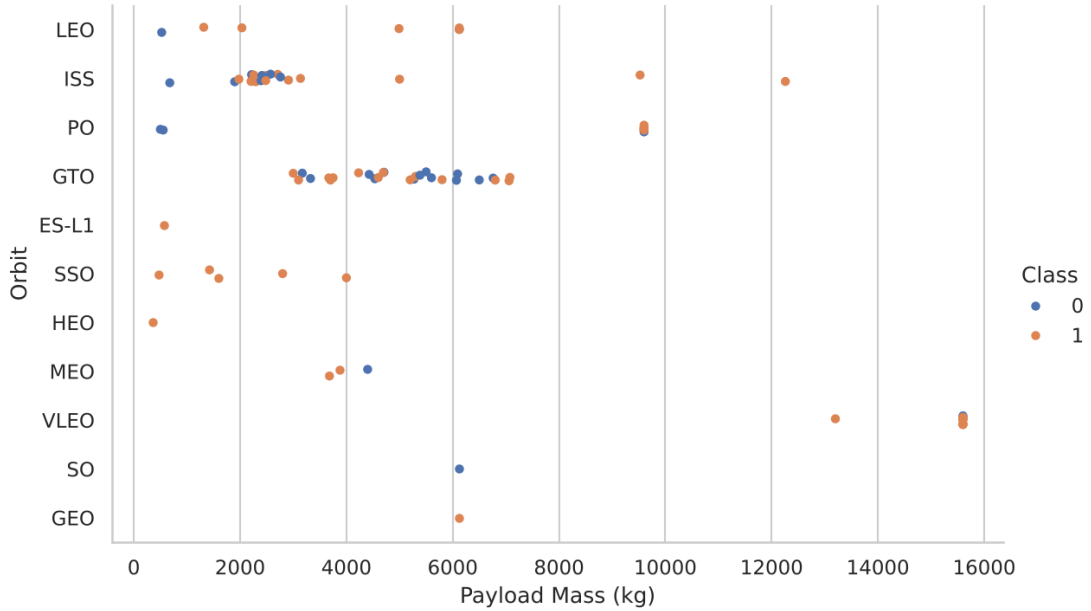


- Flights 1-45 succeed 57.8% of the time; later flights 75.6% (higher with experience).
- Payloads under 5000 kg land 63.8%; 5000 kg and above 69.8%; payloads of 9000 kg or more: 87.0% (23 launches).
- First flight number per site: CCSFS SLC 40 #1, KSC LC 39A #27, VAFB SLC 4E #4.
- Launches per site: CCSFS SLC 40 55, KSC LC 39A 22, VAFB SLC 4E 13; best success rate at KSC LC 39A.

GitHub: https://github.com/hongan151206-ai/Capstone-project

EDA Results - Payload by Site and Orbit

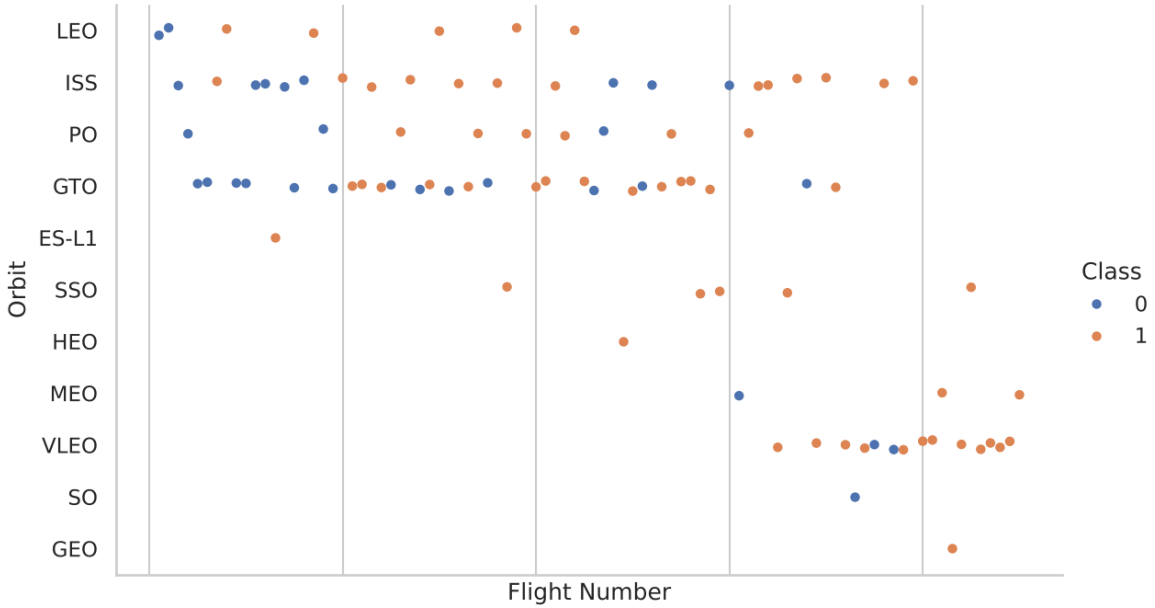


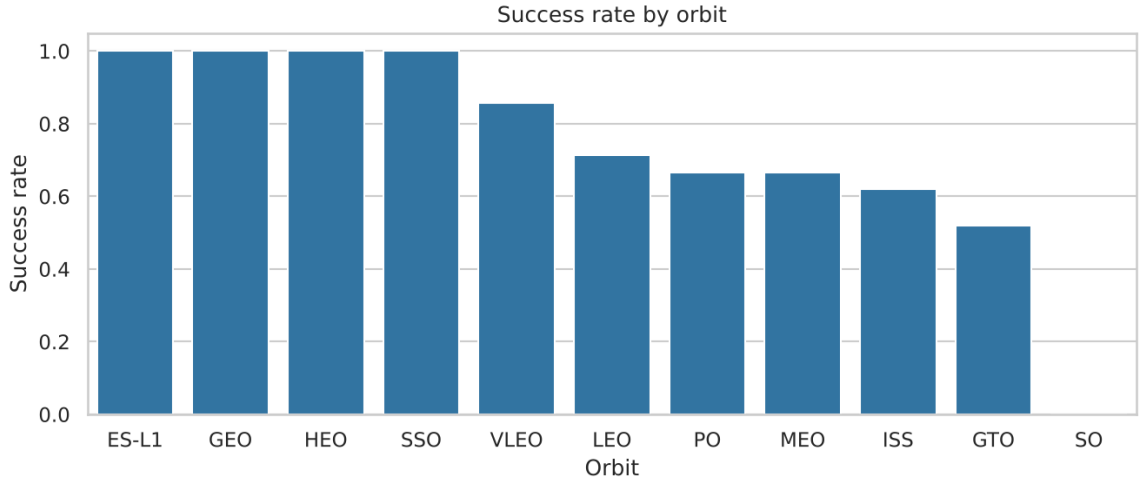


- Average payload per site: CCSFS SLC 40 5,563 kg, KSC LC 39A 7,645 kg, VAFB SLC 4E 5,919 kg.
- Orbits with the heaviest average payloads: VLEO 15,429 kg, PO 7,584 kg, GEO 6,124 kg.
- Success by site: KSC LC 39A 77.3% (22), VAFB SLC 4E 76.9% (13), CCSFS SLC 40 60.0% (55).
- Weakest site: CCSFS SLC 40; strongest: KSC LC 39A.

GitHub: https://github.com/hongan151206-ai/Capstone-project

EDA Results - Orbit Analysis

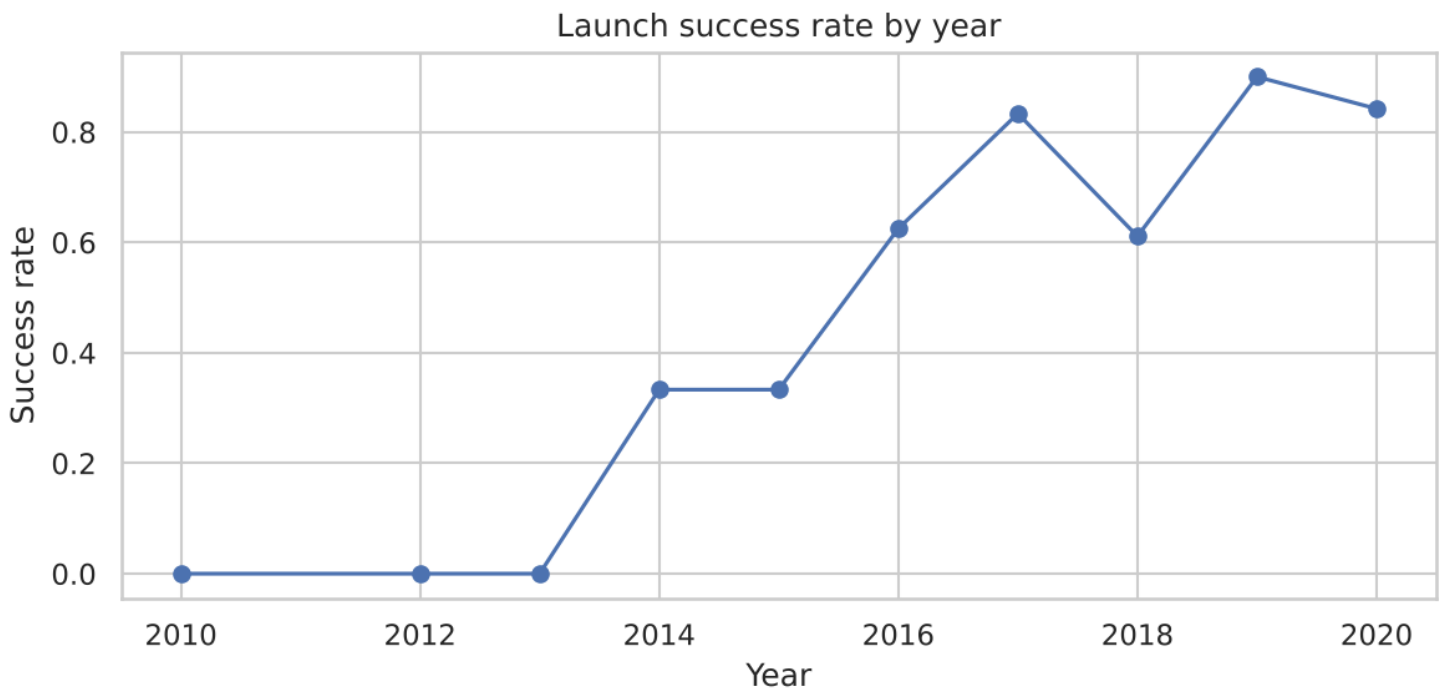




- Success rate by orbit (launches): ES-L1 100.0% (1), GEO 100.0% (1), HEO 100.0% (1), SSO 100.0% (5), VLEO 85.7% (14), LEO 71.4% (7), PO 66.7% (9), MEO 66.7% (3), ISS 61.9% (21), GTO 51.9% (27), SO 0.0% (1).
- Orbits with 100% success have very few launches (ES-L1: 1, GEO: 1, HEO: 1, SSO: 5).
- Among orbits with at least 8 launches the best is VLEO and the weakest is GTO.

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EDA Results - Yearly Trend



Success rate by year (launches)

2010: 0.0% (1)
2012: 0.0% (1)
2013: 0.0% (3)
2014: 33.3% (6)
2015: 33.3% (6)
2016: 62.5% (8)
2017: 83.3% (18)
2018: 61.1% (18)
2019: 90.0% (10)
2020: 84.2% (19)

Success rate moved from 0.0% (2010) to 84.2% (2020), peaking at 90.0% in 2019.

GitHub: https://github.com/hongan151206-ai/Capstone-project

EDA with SQL - Launch Sites and Payloads (Q1-Q4)

Q1 Unique launch sites

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Q2 Records for sites starting with 'CCA'

Date	Booster_Version	Launch_Site	Customer	Landing_Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	NASA (CRS)	No attempt

Q3 Total payload mass, NASA (CRS)

total_kg
45596

Q4 Average payload mass, booster F9 v1.1

avg_kg
2,534.7

NASA (CRS) carried 45596 kg in total; boosters F9 v1.1 averaged 2,535 kg per launch.

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EDA with SQL - Landing Success and Mission Outcomes (Q5-Q8)

Q5 First successful ground-pad landing

MIN(Date)
2015-12-22

Q6 Drone-ship success, payload 4000-6000 kg

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Q7 Successful vs failed missions

Mission_Outcome	n
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Q8 Boosters carrying the maximum payload

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2

... +4 more rows (see sql_results/q8.csv)

First successful ground-pad landing: 2015-12-22. Boosters that landed on a drone ship with 4000-6000 kg payloads and the boosters that carried the maximum payload are listed above.

EDA with SQL - Rankings and Time Analysis (Q9-Q10)

Q9 2015 failed drone-ship landings

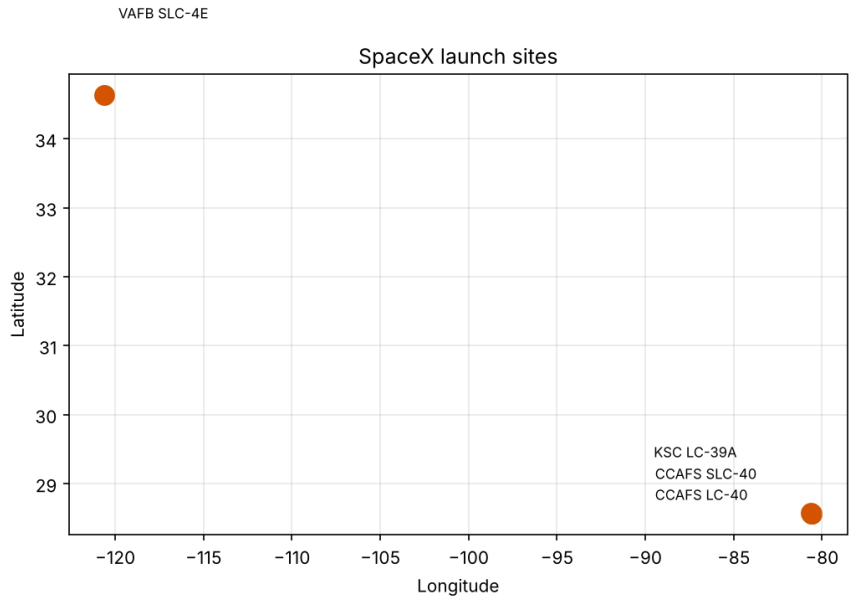
month	Booster_Version	Launch_Site
1	F9 v1.1 B1012	CCAFS LC-40
4	F9 v1.1 B1015	CCAFS LC-40

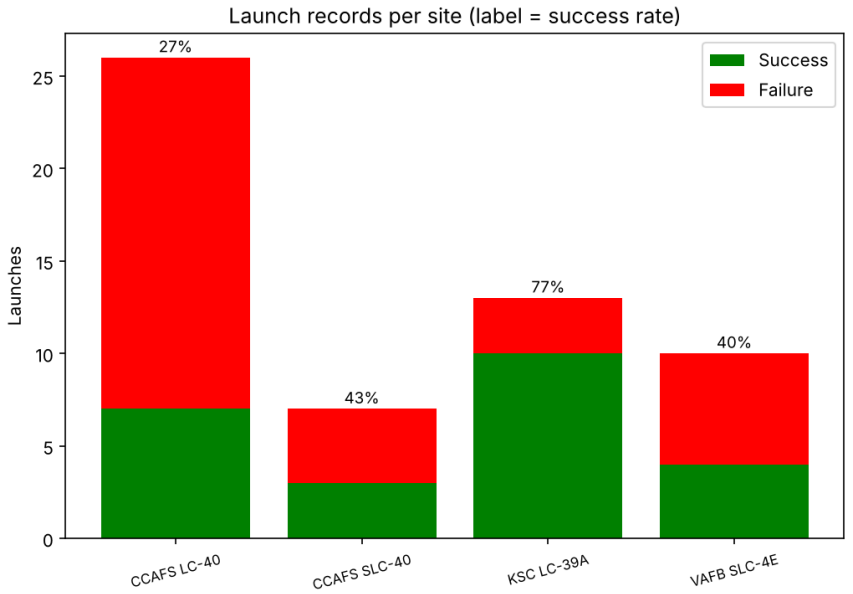
Q10 Landing outcomes ranked, 2010-06-04 to 2017-03-20

Landing_Outcome	n
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Most frequent outcome between 2010-06-04 and 2017-03-20: 'No attempt' (10 launches). Failed drone-ship landings in 2015 versus successful landings later show the landing technology maturing over time.

Folium Map - Launch Sites and Launch Records

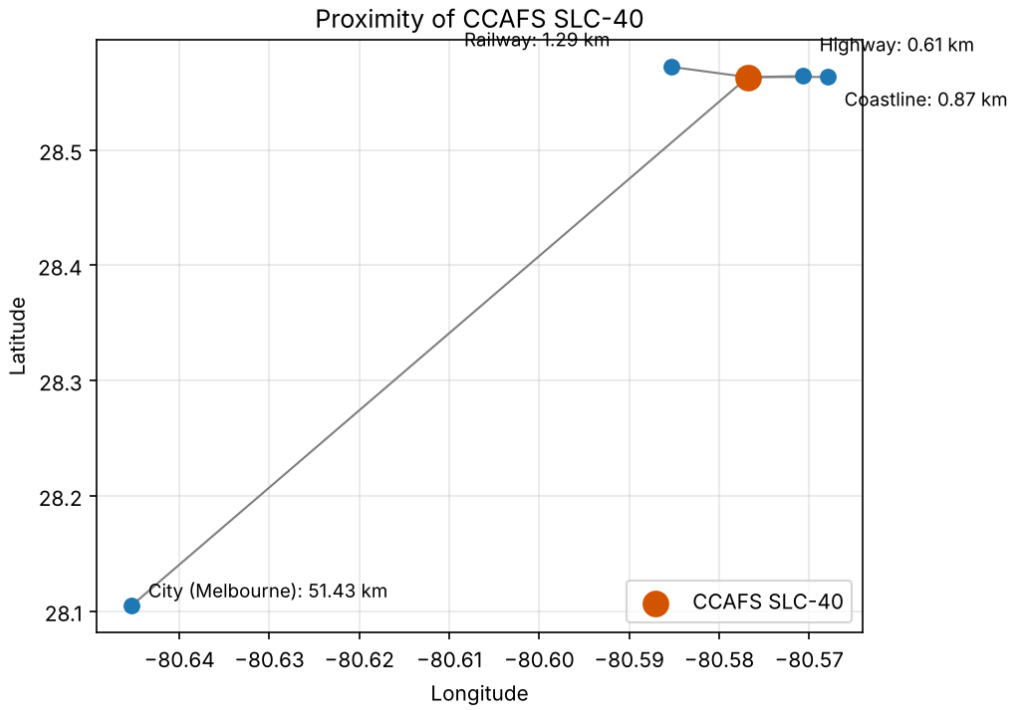




- CCAFS LC-40 (28.56, -80.58): 7 successes of 26 launch records (26.9%).
- CCAFS SLC-40 (28.56, -80.58): 3 successes of 7 launch records (42.9%).
- KSC LC-39A (28.57, -80.65): 10 successes of 13 launch records (76.9%).
- VAFB SLC-4E (34.63, -120.61): 4 successes of 10 launch records (40.0%).
- Map elements: circle + label per launch site; marker cluster of all launch records, green = success, red = failure.
- All sites are on the US coasts (Florida and California); the Florida sites are at lower latitude than VAFB SLC-4E.

GitHub: https://github.com/hongan151206-ai/Capstone-project

Folium Map - Proximity Analysis

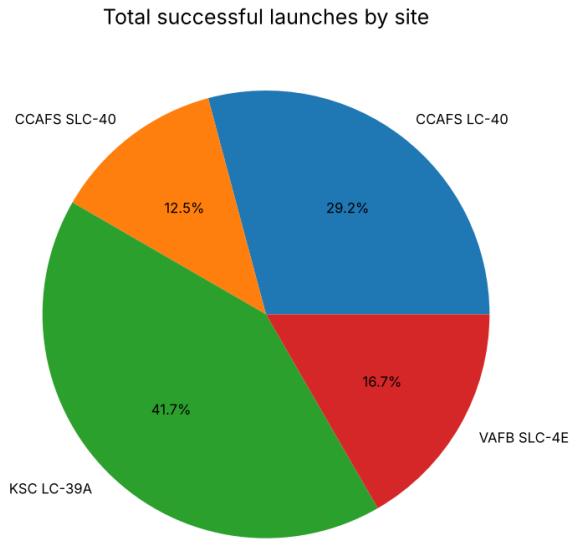


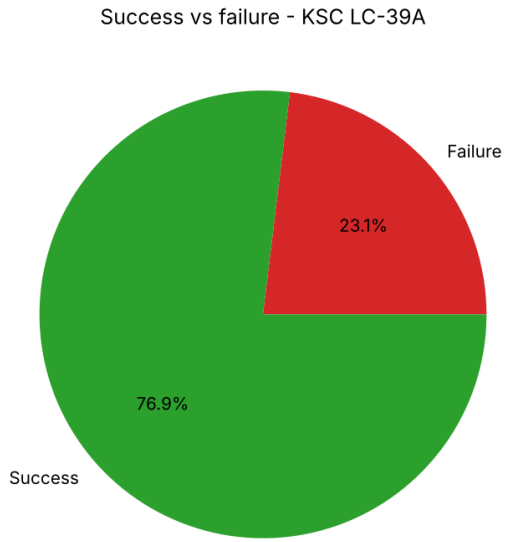
- Distance from CCAFS SLC-40 to coastline: 0.87 km
- Distance from CCAFS SLC-40 to railway: 1.29 km
- Distance from CCAFS SLC-40 to highway: 0.61 km
- Distance from CCAFS SLC-40 to city (melbourne): 51.43 km
- Method: points picked with MousePosition; distance computed with the haversine formula; PolyLine drawn to each point.
- The site is close to coast, railway and highway: easy transport of rocket parts and launches over the ocean.
- It is far from cities, which limits risk to the population.

Proximity analysis answers: are launch sites near railways, highways, coastlines? Do they keep a distance from cities? (see 06_folium_map.py)

GitHub: https://github.com/hongan151206-ai/Capstone-project

Plotly Dash - Success Pie Charts

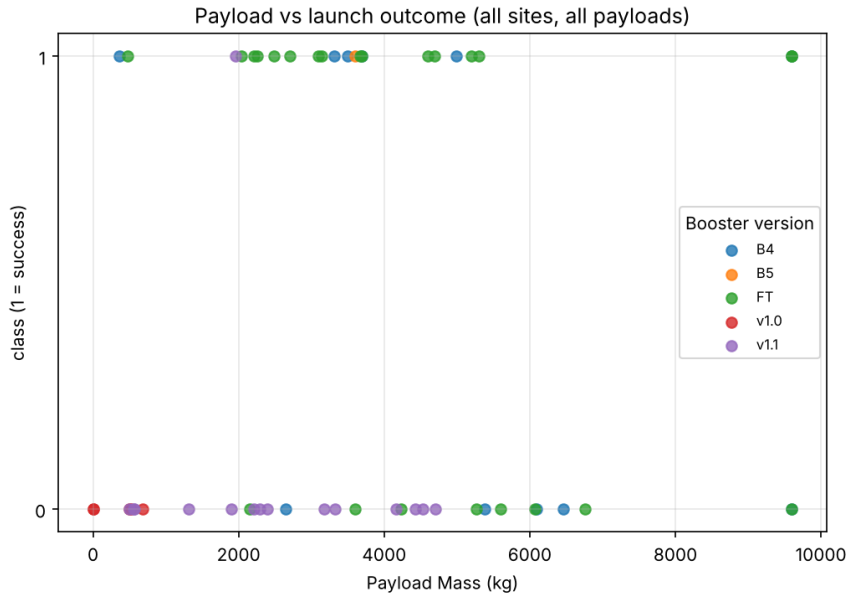


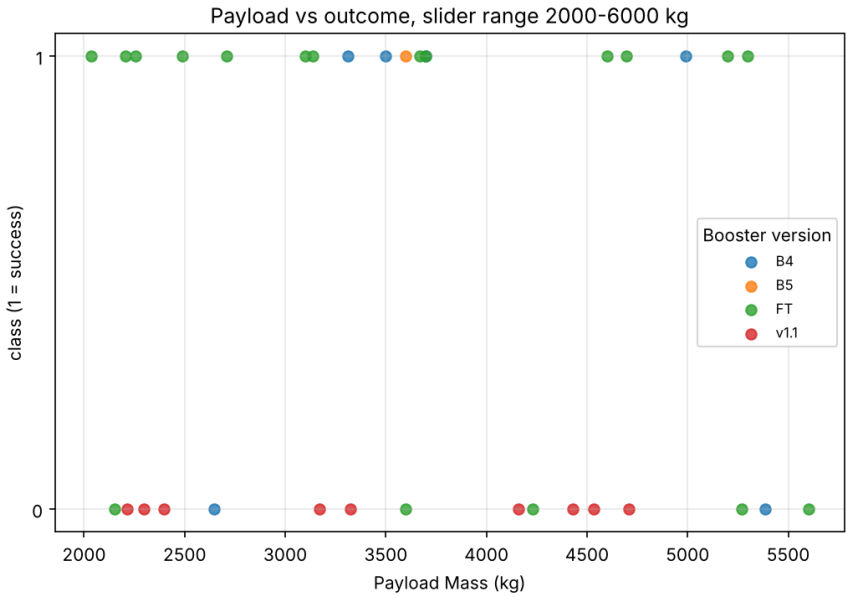


- CCAFS LC-40: 7 successful launches (29.2% of all successes), success ratio 26.9% (26 launches).
- CCAFS SLC-40: 3 successful launches (12.5% of all successes), success ratio 42.9% (7 launches).
- KSC LC-39A: 10 successful launches (41.7% of all successes), success ratio 76.9% (13 launches).
- VAFB SLC-4E: 4 successful launches (16.7% of all successes), success ratio 40.0% (10 launches).
- KSC LC-39A has the most successful launches; KSC LC-39A has the best success ratio (right pie).
- Dropdown: 'All sites' shows each site's share of successes; selecting one site shows its success vs failure split.

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Plotly Dash - Payload vs Launch Outcome

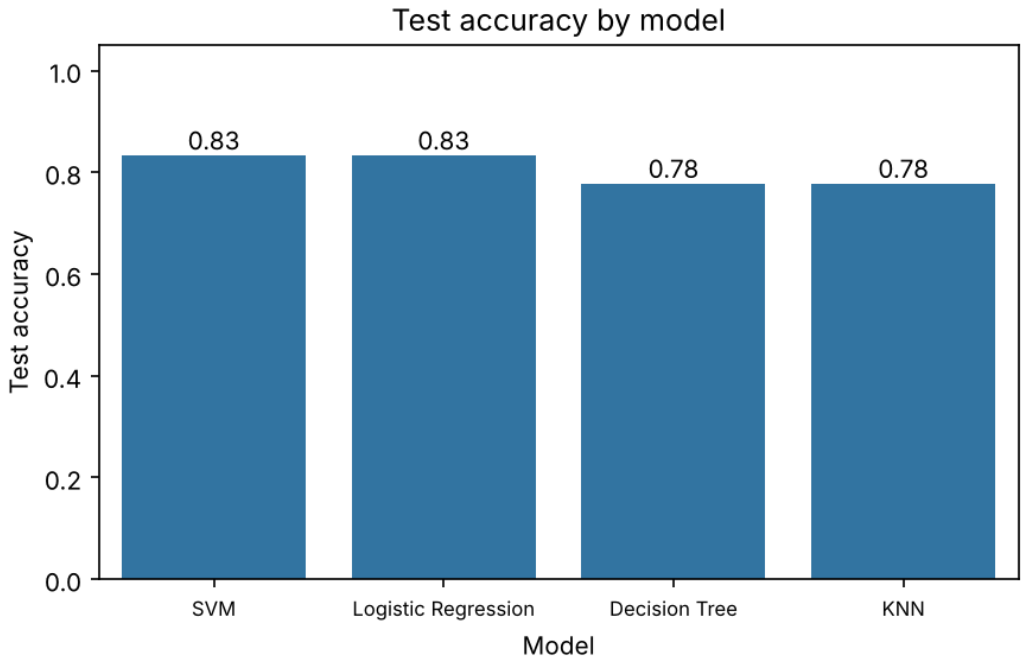




- Success rate by payload range (kg): 0-2000 23.1% (13), 2000-4000 61.9% (21), 4000-6000 38.5% (13), 6000-8000 0.0% (4), >8000 60.0% (5).
- Success rate by booster version category (launches): B4 54.5% (11), B5 100.0% (1), FT 66.7% (24), v1.0 0.0% (5), v1.1 6.7% (15).
- Payloads under 5000 kg: 45.2% success; 5000 kg and above: 35.7%.
- Range slider (right chart shows 2000-6000 kg) filters the points; colours show booster version categories.

GitHub: https://github.com/hongan151206-ai/Capstone-project

Predictive Analysis Results - Model Comparison

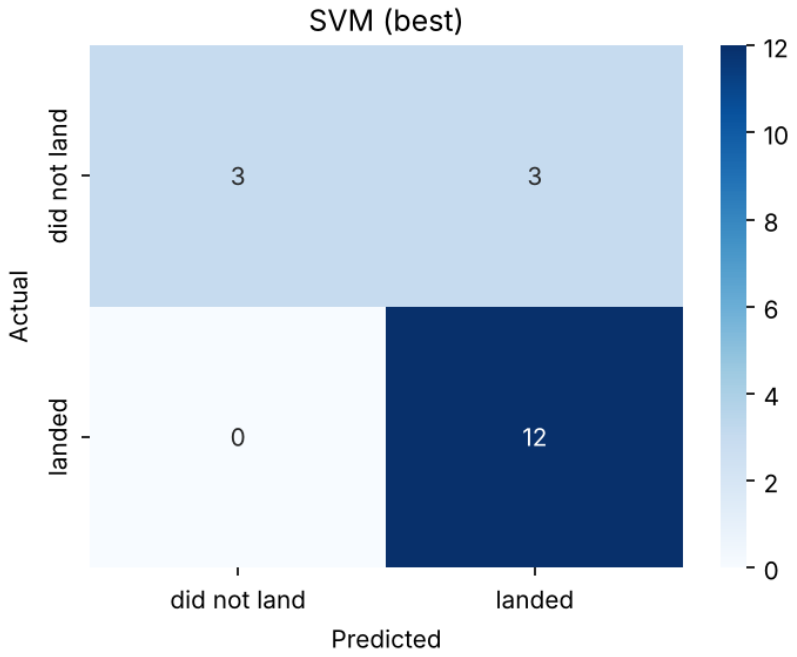


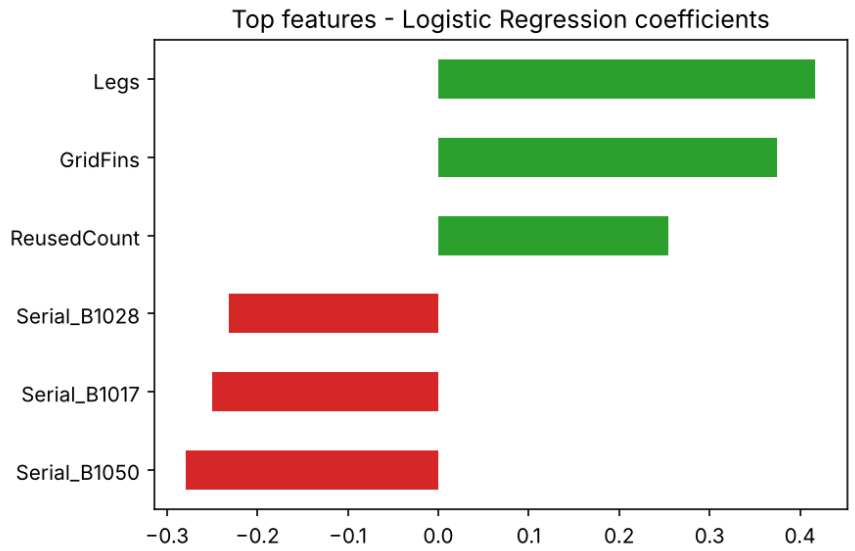
Model	CV acc	Test acc	Precision	Recall	F1
SVM	0.848	0.833	0.800	1.000	0.889
Logistic Regression	0.821	0.833	0.800	1.000	0.889
Decision Tree	0.836	0.778	0.900	0.750	0.818
KNN	0.834	0.778	0.750	1.000	0.857

- Best model: SVM - test accuracy 83.3%
- Best parameters: {'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'}

GitHub: https://github.com/hongan151206-ai/Capstone-project

Predictive Analysis Results - Confusion Matrix and Best Model





- SVM: TP 12, TN 3, FP 3, FN 0 on 18 test launches.
- FP = predicted a landing that did not happen; FN = missed a successful landing.
- Strongest features (logistic regression coefficients): Legs, GridFins, Serial_B1050, ReusedCount, Serial_B1017, Serial_B1028.

Test set is small (18 launches), so differences between models are within noise; compare the models on the F1 score and the confusion matrix as well as accuracy.

GitHub: https://github.com/hongan151206-ai/Capstone-project

Conclusion and Innovative Insights

- Landing success is predictable from pre-launch data: SVM reaches 83.3% test accuracy.
- Success changed from 0.0% to 84.2% across years; later flights succeed 75.6% of the time vs 57.8% for early ones.
- Orbit and site matter: best site KSC LC 39A; payloads below 5000 kg land 63.8% vs 69.8%.
- Innovative insight - cost model: converting predicted landing probability into expected cost (assumption: $62.0M if landed, $165.0M if not) gives about $94.8M per launch, a basis for a competing bid.
- Next steps: add weather, booster age and landing-pad data; try ensembles (Random Forest, XGBoost); validate with launches after 2020.

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